III B.Tech – II Semester

Course Code	**WATERSHED MANAGEMENT**	L	T	P	C
23A01605b	**Professional Elective-III**	3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the concept of watershed management, stakeholder roles, pollution sources, and environmental guidelines for water quality.
2. **Analyze** soil erosion processes, sediment yield, and wetland hydrology, including the role ofwaterin wetland ecosystems.
3. **Evaluate** surface water and groundwater interactions, wetland water quality, and hydrological models for effective watershed planning.
4. **Apply** principles of wetland hydrologic assessment, water harvesting, and watershed treatment system design to real-world scenarios.
5. **Assess** irrigation planning, participatory water management, and water footprint concepts to ensure sustainable water resource utilization.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. **Explain** watershed management concepts, pollution control strategies, and environmental policies related to water quality.
2. **Analyze** erosion processes, wetland water budgets, and sediment transport models to assess land degradation and conservation needs.
3. **Evaluate** surface and groundwater interactions, wetland treatment efficiency, and hydrological models for integrated water resource management.
4. **Apply** water harvesting techniques, hydrologic modeling, and wetland design methods for sustainable watershed management.
5. **Assess** irrigation water management strategies, drought mitigation policies, and the role of water footprintin agricultural sustainability.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	1	2	3	1	-	2	-	2	2	2
CO -2	3	3	2	3	2	2	3	1	-	2	-	2	2	2
CO -3	3	3	3	3	3	2	3	2	-	2	-	2	3	3
CO -4	3	2	3	3	3	3	3	2	-	2	2	2	3	3
CO -5	3	3	3	3	3	3	3	2	-	2	2	2	3	3

UNIT – I

Concept of Watershed, Introduction to Watershed Management, Different Stakeholders and Their Relative Importance, Watershed Management Policies and Decision Making, Watershed Management Practices in Arid and Semiarid Regions, Short Term and Long Term Strategic Planning, Types and Sources of Pollution, Environmental Guidelines for Water Quality, Perspective On Recycle and Reuse

UNIT – II

Morphometry, Soil Erosion - Erosion - Factors Affecting Erosion, Effects of Erosion On Land Fertility and Land Capability, Soil Erosion Modelling, Erosivity and Erodibility - Sediment Yield and Sedimentation- Wetland Definitions and The Role of Water in Wetland Structure

and Function, Introduction to Wetland Water Budgets and Hydro-Period Components of The Water Budget: Inflows, Outflows, and Storage, Precipitation and Runoff, Evapotranspiration;

UNIT – III

Surface Water Flows: Structures and Channels, Groundwater-Surface Water Exchange inWetlands, Surface Water Flows Ii and Wetland Hydrology Case Studies, Flow and Mixing in Wetlands Wetland Water Quality Information: Nutrients, Organic/Inorganic Contaminants, Sediments and Colloids, Wetland Transport Models I: Plug Flow, Cstrs andCstrs in Series; Intro to Method of Moments.

UNIT – IV

Wetland Hydrologic Assessment: Physical and Biological Processes, Anthropogenic and Climate Change Impacts On Wetland Hydrology, Modeling Wetland Hydrology, Hydraulics, and Hydrodynamics, Introduction to Wetland Treatment Systems Design - Water Harvesting: Rainwater Harvesting, Catchment Harvesting, Harvesting Structures - Model Watershed – Government and Ngo Projects.

UNIT – V

Rain Water Management. Planning and Operation of Irrigation Systems. Conjunctive Use of Water. Participatory Irrigation Management and Integrated Water Resources Management (IWRM), Water Management Policy During Droughts. Predicting Effect of Water Shortage On Crops. Introduction to Water Footprint of Crops and Its Applications. Blue, Green and Grey Water Foot Print.

TEXT BOOKS:

1. T. O. Randhir, Watershed Management: Issues and Approaches, IWA Publishing, 2006
2. J. V. S. Murty, Watershed Management, New Age International, 2013

REFRENCE BOOKS:

1. D. K. Majumdar, Irrigation Water Management, Prentice Hall, 2014
2. K. N. Brooks, P. F. Folliott, J. A. Magner, Hydrology and the Management of Watersheds, Wiley-Blackwell, Fourth edition, 2012
3. E. M. Tideman, Watershed Management: Guidelines for Indian Conditions, Omega Scientific Publishers, 1996
4. R. Rajora, Integrated Watershed Management: Field Manual for Equitable, Productive and Sustainable Development, Rawat Publications, 2019

Online Learning Resources:

https://nptel.ac.in/courses/105101010
https://nptel.ac.in/courses/126105334